

Chapter 19: Basis and Coordinates

Theorem: If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a linearly independent set of vectors in V , then every vector in $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ has a unique representation as a linear combination of the $\mathbf{v}_i$.

Proof: Let $\vec{u} = a_1 \vec{v}_1 + \dots + a_n \vec{v}_n = b_1 \vec{v}_1 + \dots + b_n \vec{v}_n$

$$\Rightarrow (a_1 - b_1) \vec{v}_1 + \dots + (a_n - b_n) \vec{v}_n = \vec{0}.$$

$$\Rightarrow a_1 = b_1, \dots, a_n = b_n.$$

Definition: The vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ in a vector space V are said to form a **basis** for V if

1. $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ **span** V , and
2. $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are **linearly independent**.

If a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ forms a basis for V , then *every* vector can be written as a *unique* linear combination of these vectors.

Example: Let $V = \mathbb{R}^3$. The vectors

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

form a basis for $\mathbb{R}^3$. What did we call this basis?

$$\textcircled{1} \quad \forall \vec{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in \mathbb{R}^3, \quad \vec{v} = a \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

$$\textcircled{2} \quad a_1 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + a_2 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + a_3 \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow a_1 = a_2 = a_3, \text{ so they are L.I.}$$

So it is a basis of $\mathbb{R}^3$.

Fact: The set of vectors $\{x^d, x^{d-1}, \dots, x^2, x, 1\}$ forms a basis for the vector space P_d called the **standard basis**, or the **natural basis**.

Example: Show that the set $\{x+1, x-1\}$ forms a basis for P_1 .

$$\textcircled{1} \quad a_1 \cdot (x+1) + a_2 \cdot (x-1) = 0 \Rightarrow a_1 = a_2 = 0,$$

so the set is L.I.

$$\textcircled{2} \quad \forall p(x) = ax + b \in P_1, \quad p(x) = a \cdot (x+1) + \frac{(b-a)}{2} \cdot [(x+1) - (x-1)]$$

$$= \frac{a+b}{2}(x+1) + \frac{a-b}{2}(x-1)$$

so $p(x) \in \text{Span}\{x+1, x-1\}$

Coordinates

so $\{x+1, x-1\}$ is a basis for P_1 .

Let V be a vector space of dimension n , and fix an ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ for V . Let $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$.

Let $\mathbf{v}$ be a vector in V . Then $\mathbf{v}$ can be *uniquely expressed* in the form

$$\mathbf{v} = a_1 \mathbf{b}_1 + a_2 \mathbf{b}_2 + \dots + a_n \mathbf{b}_n,$$

where $a_1, a_2, \dots, a_n$ are real numbers.

So we can build a coordinate representation of $\mathbf{v}$:

$$C_B(\mathbf{v}) = C_B(a_1 \mathbf{b}_1 + a_2 \mathbf{b}_2 + \dots + a_n \mathbf{b}_n) = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + \dots + a_n \mathbf{e}_n = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

$C_B(\mathbf{v})$ is called the **coordinate vector** in $\mathbb{R}^n$ for $\mathbf{v}$ relative to the basis B , realized as a column vector. Your book also denotes this as $[\mathbf{v}]_B$.

Remark: Your book also defines $(\mathbf{v})_B = (a_1, \dots, a_n)$ to mean the coordinate vector in $\mathbb{R}^n$ for $\mathbf{v}$ relative to the basis B , realized as a row vector. We will not work with this vector.

Example: For a given basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$, what is $C_B(\mathbf{b}_i)$?

$$\vec{b}_i = 0 \cdot \vec{b}_1 + \dots + 1 \cdot \vec{b}_i + \dots + 0 \cdot \vec{b}_n$$

$$\Rightarrow C_B(\vec{b}_i) = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} - i^{\text{th}}$$

Warning: The order of the elements in B matters!

Example: Let $B_1 = (1, x, x^2)$ be a basis for P_2 , and $p = x^2 + 2x - 3$. What is $C_{B_1}(p)$?

$$p(x) = -3 \cdot 1 + 2 \cdot x + 1 \cdot x^2.$$

$$C_{B_1}(p) = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$$

Example: Let $B_2 = (x, x^2, 1)$ be a basis for P_2 , and $p = x^2 + 2x - 3$. What is $C_{B_2}(p)$?

$$C_{B_2}(p) = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$$

Example: Let $B = (x+1, x-1)$ be a basis for P_1 . If $C_B(p) = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$, what is the polynomial p ?

$$\begin{aligned} p(x) &= 3 \cdot (x+1) - 2(x-1) \\ &= x + 5. \end{aligned}$$

Example: Let $B = \left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right)$. You can verify this is a basis for the set of 2×2 symmetric matrices. If $C_B(A) = \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}$, find A .

$$A = -3 \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + 4 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -3 & 0 \end{bmatrix}$$

Chapter 20: Dimension

We've seen that a basis for a vector space gives a way to represent each vector as a unique linear combination of the basis. A basis is not necessarily unique; however, we will now show that the number of vectors in a basis is unique.

Theorem: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ spans a vector space V , and $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ is a set of linearly independent vectors in V , then $m \leq n$.

Sketch of proof: Claim: Any $n+1$ vectors are L.D.

Let $\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_{n+1}\}$ be arbitrary $n+1$ vectors.
 Since S spans V , $\Rightarrow \vec{u}_1 = a_{11} \cdot \vec{v}_1 + \dots + a_{1n} \cdot \vec{v}_n$
 $\vdots$
 $\vec{u}_{n+1} = a_{n+1,1} \cdot \vec{v}_1 + \dots + a_{n+1,n} \cdot \vec{v}_n$

Let $k_1 \vec{u}_1 + \dots + k_{n+1} \vec{u}_{n+1} = \vec{0} \Rightarrow k_1 (a_{11} \vec{v}_1 + \dots + a_{1n} \vec{v}_n) + \dots + k_{n+1} (a_{n+1,1} \vec{v}_1 + \dots + a_{n+1,n} \vec{v}_n) = \vec{0}$
 $\Rightarrow (a_{11} k_1 + a_{21} k_2 + \dots + a_{n+1,1} k_{n+1}) \vec{v}_1 + \dots + (a_{1n} k_1 + \dots + a_{n+1,n} k_{n+1}) \vec{v}_n = \vec{0}$
 So $k_1, \dots, k_{n+1}$ have nonzero s.d., so L.D.

This proof also shows that m of the spanning vectors in S can be replaced by $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$, and the new set will still span V .

Theorem: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ are both bases for a vector space V , then $m = n$.

Sketch of proof: By the previous thm, $m \leq n$, also $n \leq m \Rightarrow m = n$.

Definition: The **dimension** of a nonzero vector space V is the number of vectors in a basis for V . We often write $\dim V$ for the dimension of V . By convention, the dimension of the trivial vector space $\{\mathbf{0}\}$ is 0.

Example: What is the dimension of P_2 ? Of P_3 ? Of P_d ?

$$\dim P_2 = 3, \quad \dim P_3 = 4,$$

$$\dim P_d = d+1.$$

Example: What is the dimension of M_{23} ? Of M_{mn} ?

$$\dim M_{2 \times 3} = 6$$

Example: Let W be the subspace of 2×2 symmetric matrices in M_{22} . Find the dimension of W .

$$\dim W = 3.$$

Definition: A vector space V is called **finite dimensional** if it is spanned by a finite set of vectors. Otherwise, V is called **infinite dimensional**.

Example: Let P be the set of all polynomials. Is it possible that a finite set spans P ?

NO, it is infinite dimensional
vector space

Example: Is $\{\mathbf{0}\}$ finite dimensional?

Yes, $\dim \{\mathbf{0}\} = 0.$

Lemma: Let V be a finite dimensional vector space. If U is any subspace of V , then any linearly independent subset of U can be enlarged to a finite basis of U .

Proof: If U spans V , then we're done. If NOT, then

adding new vectors to U until U contains n L.I. vectors.

Using this lemma, we can build a basis for any finite dimensional vector space...

Theorem: Let V be a finite dimensional vector space spanned by m vectors. Assume $V \neq \{0\}$.

1. V has a finite basis, and $\dim V \leq m$.
2. Every linearly independent set of vectors in V can be enlarged to a basis of V by adding vectors from any fixed basis of V .
3. If U is a subspace of V , then
 - a. U is finite dimensional and $\dim U \leq \dim V$.
 - b. If $\dim U = \dim V$ then $U = V$.

Theorem: Let V be a finite dimensional vector space. Assume $V \neq \{0\}$. Any spanning set S for V can be cut down (by deleting vectors) to a basis of V .

Idea of proof: If S is L.I., we're done. If not, then pick the vector that is a linear comb. of others, and delete it, until the # of vectors in S is n .

Example: Find the dimension of the subspace W of $\mathbb{R}^4$ spanned by $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 5 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 3 \\ 0 \\ 1 \\ -2 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -5 \\ 4 \\ 9 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{cases} a_1 \vec{v}_1 + a_2 \vec{v}_2 + a_3 \vec{v}_3 = \vec{0} \\ \Rightarrow \begin{bmatrix} -1 & 3 & -5 \\ 2 & 0 & 4 \\ 5 & 1 & 9 \\ 0 & -2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{cases}$$

$$\Rightarrow \begin{cases} a_1 = -2a_3 \\ a_2 = a_3 \end{cases}$$

(More space for example:) $\Rightarrow \begin{cases} a_1 = -2 \\ a_2 = 1 \\ a_3 = 1 \end{cases}$ is a nonzero sol'n.

$$\Rightarrow -2 \cdot \vec{v}_1 + 1 \cdot \vec{v}_2 + 1 \cdot \vec{v}_3 = \vec{0} \Rightarrow \vec{v}_3 = 2\vec{v}_1 - \vec{v}_2.$$

$\Rightarrow \vec{v}_3$ is redundant. $\{\vec{v}_1, \vec{v}_2\}$ is a basis.

So $\dim W = 2$.

Theorem: Let V be an n -dimensional vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a set of vectors in V .

1. If S is linearly independent, then S is a basis for V .
2. If S spans V , then S is a basis for V .

By the theorem above, if we know the dimension of V , to check if a set S with $\dim V$ vectors is a basis for V , we only need to check one of linear independence or spanning.

Example: Let W denote the space of all symmetric 2×2 matrices. Find a basis of W consisting of invertible matrices.

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}.$$

S spans W , and S contain 3 L.I. invertible matrices, by the previous thm, S is a basis.